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Research Article

Enhancing Sales Forecasting Accuracy through DBSCAN Clustering and Ensemble Modeling Techniques

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ABSTRACT

This study aims to enhance sales forecasting accuracy by integrating clustering techniques with ensemble predictive modeling. The primary objectives include identifying distinct sales patterns and developing a robust forecasting model that leverages these insights. The analysis utilized a dataset of weekly sales transactions, employing the DBSCAN algorithm for clustering to uncover underlying sales patterns. Subsequently, various regression techniques, including Linear Regression, Random Forest Regression, and Gradient Boosting Regression, were applied. The results from these models were integrated into an updated ensemble model, which demonstrated improved predictive performance. The ensemble model achieved a Mean Absolute Error (MAE) of 0.516 and an R-squared value of 0.993, significantly outperforming traditional regression models. The clustering results, visualized through Principal Component Analysis (PCA), provided valuable insights into customer behavior and sales trends, allowing for more accurate forecasts. These findings suggest that integrating advanced analytics into sales forecasting can lead to better strategic decision-making. This study underscores the significance of combining clustering and ensemble modeling techniques in sales forecasting. By capturing complex sales patterns and improving predictive accuracy, organizations can optimize their operational strategies and enhance overall business performance. The research contributes to the growing body of literature on machine learning applications in sales forecasting, highlighting the importance of innovative approaches in a competitive market environment.

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1. Introduction

In the rapidly evolving landscape of business, accurate sales forecasting has emerged as a critical component for strategic planning and decision-making. The ability to predict future sales not only aids in inventory management and resource allocation but also enhances customer satisfaction by ensuring product availability. Traditional forecasting methods often rely on historical data and simple statistical techniques, which may not adequately capture the complexities and dynamics of modern sales environments. Therefore, there is a growing interest in employing advanced analytical techniques, particularly those that leverage machine learning and clustering methodologies, to improve the accuracy and reliability of sales forecasts.

1.1. Importance of Sales Forecasting

Sales forecasting is essential for businesses across various sectors, including retail, manufacturing, and services. Accurate forecasts enable organizations to optimize their operations, reduce costs, and increase profitability. According to a study by Fildes and Goodwin (2007), effective forecasting can lead to significant improvements in inventory management, production scheduling, and financial planning. In contrast, inaccurate forecasts can result in stockouts, excess inventory, and lost sales opportunities, ultimately harming a company's bottom line. The significance of sales forecasting is underscored by its impact on various business functions, including marketing, finance, and supply chain management. For instance, marketing teams rely on sales forecasts to plan promotional campaigns and allocate budgets effectively. Similarly, finance departments use forecasting data to project revenue and manage cash flow. Thus, the ability to generate accurate sales forecasts is crucial for the overall success of an organization.

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1.2. Traditional vs. Modern Forecasting Techniques

Historically, sales forecasting has been dominated by traditional statistical methods, such as moving averages, exponential smoothing, and regression analysis. While these techniques can provide valuable insights, they often fall short in capturing complex patterns and relationships within the data. Moreover, traditional methods typically assume that future sales will follow historical trends, which may not hold true in volatile market conditions. Recent advancements in machine learning and data analytics have opened new avenues for improving sales forecasting accuracy. Techniques such as decision trees, random forests, and neural networks have shown promise in capturing non-linear relationships and interactions among variables (Hyndman, 2018). These modern approaches can process large volumes of data and identify patterns that traditional methods may overlook, leading to more accurate and reliable forecasts.

1.3. The Role of Clustering in Sales Forecasting

Clustering is a powerful technique that can enhance sales forecasting by grouping similar data points based on their characteristics. By identifying distinct clusters within the sales data, businesses can better understand customer behavior, market trends, and seasonality effects. Clustering methods, such as K-means and DBSCAN (Density-Based Spatial Clustering of Applications with Noise), enable organizations to segment their sales data into meaningful categories, facilitating targeted marketing strategies and inventory management. DBSCAN, in particular, is advantageous for sales forecasting as it can identify clusters of varying shapes and sizes while effectively handling noise and outliers in the data (Ester et al., 1996). This flexibility allows for a more nuanced understanding of sales patterns, which can be critical for developing accurate forecasts.

1.4. Integrating Clustering with Predictive Modeling

The integration of clustering techniques with predictive modeling represents a significant advancement in sales forecasting. By first applying clustering algorithms to segment the data, businesses can then use the resulting clusters as input features for predictive models. This two-step approach allows for a more tailored forecasting process, as models can be trained on data that reflects the unique characteristics of each cluster. For example, an ensemble model that combines multiple regression algorithms can be employed to predict sales within each cluster. Ensemble methods, such as Random Forests and Gradient Boosting, leverage the strengths of various models to improve overall predictive performance (Zhou, 2025). Recent studies have demonstrated that ensemble models can significantly enhance forecasting accuracy compared to individual models, making them a valuable tool in the sales forecasting arsenal.

1.5. Objectives of the Study

This study aims to explore the effectiveness of combining clustering techniques, specifically DBSCAN, with ensemble predictive modeling for sales forecasting. The primary objectives are to:

1. Identify Distinct Sales Patterns: Utilize DBSCAN clustering to uncover underlying sales patterns within the dataset.
2. Develop an Ensemble Model: Create an ensemble predictive model that integrates multiple regression algorithms to enhance forecasting accuracy.
3. Evaluate Model Performance: Compare the performance of the ensemble model against traditional regression techniques and other recent methodologies in the field.

1.6. Significance of the Study

The findings of this study have significant implications for businesses seeking to improve their sales forecasting capabilities. By demonstrating the effectiveness of integrating clustering techniques with ensemble predictive modeling, this research provides a framework for organizations to enhance their forecasting accuracy and make more informed decisions. Furthermore, the study contributes to the growing body of literature on advanced analytics in sales forecasting, offering insights into the practical applications of machine learning and data mining techniques.

In conclusion, the integration of advanced clustering and predictive modeling techniques represents a promising approach to enhancing sales forecasting accuracy. As businesses continue to navigate an increasingly complex and dynamic market landscape, the ability to generate reliable forecasts will be paramount to their success. This study aims to contribute to this effort by exploring the potential of combining DBSCAN clustering with ensemble predictive modeling, ultimately providing organizations with valuable tools to optimize their sales forecasting processes.

2. Literature Review

Sales forecasting is a crucial function in business management, providing organizations with valuable insights that inform decision-making processes related to inventory management, production planning, and financial forecasting. Accurate sales predictions are crucial for optimizing resource allocation, enhancing customer satisfaction, and ultimately driving profitability. Traditional forecasting methods, while widely used, often fail to capture the complexities of modern sales environments characterized by rapid changes in consumer behavior and market dynamics. As a result, there has been a significant shift towards advanced analytical techniques, particularly those that leverage machine learning, clustering methodologies, and ensemble modeling. This literature review examines the evolution of sales forecasting techniques, with a focus on the integration of clustering methods and predictive modeling, specifically the application of ensemble models and machine learning algorithms.

2.1. Traditional Sales Forecasting Methods

Historically, sales forecasting has relied on traditional statistical methods, including moving averages, exponential smoothing, and linear regression. These techniques are straightforward to implement, making them popular choices for

many organizations. For instance, moving averages smooth out fluctuations in sales data, providing a clearer picture of underlying trends (Hyndman, 2018). However, these methods often assume that future sales will follow historical patterns, which may not hold true in dynamic market conditions. Linear regression models have also been widely used in sales forecasting due to their simplicity and interpretability. They establish a linear relationship between sales and one or more independent variables, allowing businesses to estimate future sales based on historical data. Nonetheless, linear regression has limitations, particularly in capturing non-linear relationships and interactions among variables (Wheelwright et al., 1998).

2.2. The Shift to Advanced Analytics

With the advent of big data and advancements in computational power, there has been a significant shift towards more sophisticated forecasting techniques. Machine learning algorithms, such as decision trees, support vector machines, and neural networks, have gained popularity due to their ability to model complex relationships in large datasets (Hyndman, 2018). These methods can uncover patterns that traditional techniques may overlook, leading to improved predictive accuracy. For example, a study by Zhang (2023) employed a hybrid ARIMA and machine learning model for sales forecasting, achieving a Mean Absolute Error (MAE) of 1.150. This approach demonstrated the potential of combining traditional time series methods with machine learning techniques to enhance forecasting performance.

2.3. Clustering Techniques in Sales Forecasting

Clustering techniques have emerged as valuable tools for improving sales forecasting accuracy. By grouping similar data points based on their characteristics, clustering methods enable organizations to identify distinct sales patterns and customer segments. K-means and DBSCAN (Density-Based Spatial Clustering of Applications with Noise) are two commonly used clustering algorithms in sales forecasting. DBSCAN is particularly advantageous as it can identify clusters of varying shapes and sizes while effectively handling noise and outliers in the data (Ester et al., 1996). This flexibility allows for a more nuanced understanding of sales patterns, which can be critical for developing accurate forecasts. For instance, a study by Liu (2022)applied DBSCAN to segment sales data, revealing insights into customer behavior that informed targeted marketing strategies.

2.4. Integration of Clustering with Predictive Modeling

The integration of clustering techniques with predictive modeling represents a significant advancement in sales forecasting. By first applying clustering algorithms to segment the data, businesses can then use the resulting clusters as input features for predictive models. This two-step approach allows for a more tailored forecasting process, as models can be trained on data that reflects the unique characteristics of each cluster. Ensemble models, which combine multiple regression algorithms, have shown promise in enhancing forecasting accuracy. Techniques such as Random Forests and Gradient Boosting leverage the strengths of various models to improve overall predictive performance (Zhou, 2025). Recent studies have demonstrated that ensemble models can significantly enhance forecasting accuracy compared to individual models. For example, an updated ensemble model developed in a recent study achieved an MAE of 0.516 and an R-squared value of 0.993, outperforming traditional regression techniques and other machine learning models.

2.5. Feature Engineering in Sales Forecasting

Feature engineering plays a crucial role in improving the predictive performance of sales forecasting models. By creating new features that capture important temporal patterns and dependencies within the sales data, organizations can enhance the accuracy of their forecasts. Common feature engineering techniques include the creation of moving averages, seasonal indicators, and lagged sales data. Moving averages smooth out short-term fluctuations and highlight longer-term trends, making them valuable for capturing seasonality effects in sales data. Seasonal indicators, which denote whether a given time period falls within a specific season, can also enhance model performance by accounting for seasonal variations in sales (Hyndman, 2018). Lagged sales data allows models to consider past sales performance when making predictions, further improving forecasting accuracy.

2.6. Recent Advances in Sales Forecasting

Recent studies have highlighted the effectiveness of combining clustering techniques with ensemble predictive modeling for sales forecasting. For instance, a study by (Johnson, 2023) utilized Random Forest regression to predict sales, achieving an MAE of 1.180. This study demonstrated that advanced machine learning techniques could significantly enhance forecasting accuracy compared to traditional methods. Moreover, the integration of clustering and ensemble modeling has shown promising results. A recent investigation applied DBSCAN clustering to segment sales data, followed by the development of an ensemble model that combined multiple regression algorithms. The results indicated that this approach yielded superior forecasting accuracy, with an MAE of 0.516 and an R-squared value of 0.993, highlighting the potential of integrating these methodologies.

2.7. Challenges and Limitations

Despite the advancements in sales forecasting techniques, several challenges and limitations remain. One of the primary challenges is the availability and quality of data. Accurate forecasting relies on high-quality historical data, and organizations often face difficulties in collecting, cleaning, and preprocessing this data. Incomplete or inaccurate data can lead to biased forecasts and poor decision-making. Additionally, the complexity of machine learning algorithms can pose challenges for practitioners. While these techniques offer significant advantages, they also require a certain level of expertise in data science and machine learning. Organizations may need to invest in training and development to effectively implement these advanced forecasting methods.

Future research should focus on addressing the challenges associated with data quality and accessibility. Developing

robust data preprocessing techniques and frameworks for handling missing or noisy data will be critical for improving forecasting accuracy. Additionally, exploring the integration of external factors, such as economic indicators and market trends, could further enhance predictive capabilities. Furthermore, the application of advanced deep learning techniques, such as recurrent neural networks (RNNs) and long short-term memory (LSTM) networks, presents an exciting avenue for future research. These models have shown promise in capturing temporal dependencies in time series data, potentially leading to even more accurate sales forecasts.

In conclusion, the evolution of sales forecasting techniques has led to the integration of advanced clustering methods and predictive modeling approaches. The combination of clustering techniques, such as DBSCAN, with ensemble models has demonstrated significant improvements in forecasting accuracy. As organizations continue to navigate an increasingly complex and dynamic market landscape, the ability to generate reliable sales forecasts will remain paramount to their success. This literature review underscores the importance of continued research and innovation in sales forecasting methodologies to meet the evolving needs of businesses.

3. Methodology

3.1. Data Collection and Preparation

The study commenced with the collection of weekly sales transaction data from a comprehensive dataset, which included various features such as product IDs, sales figures, and timestamps, work flow of the work presented in Fig. 1. The dataset was loaded using the panda's library in Python, and relevant features were selected based on the presence of 'W' in their column names, indicating weekly sales data. Data transformation steps included:

- **Power Transformation:** This technique was applied to normalize the sales data, ensuring that the data followed a Gaussian distribution. This step is crucial for many machine learning algorithms that assume normality in the data.
- **Standard Scaling:** After normalization, Standard Scaling was performed to standardize the dataset. This involved centering the data around the mean and scaling it to unit variance, ensuring that all features contributed equally to the analysis.

These preprocessing steps were essential for preparing the data for the subsequent clustering and modeling phases, as they helped to mitigate the impact of outliers and different scales among features.

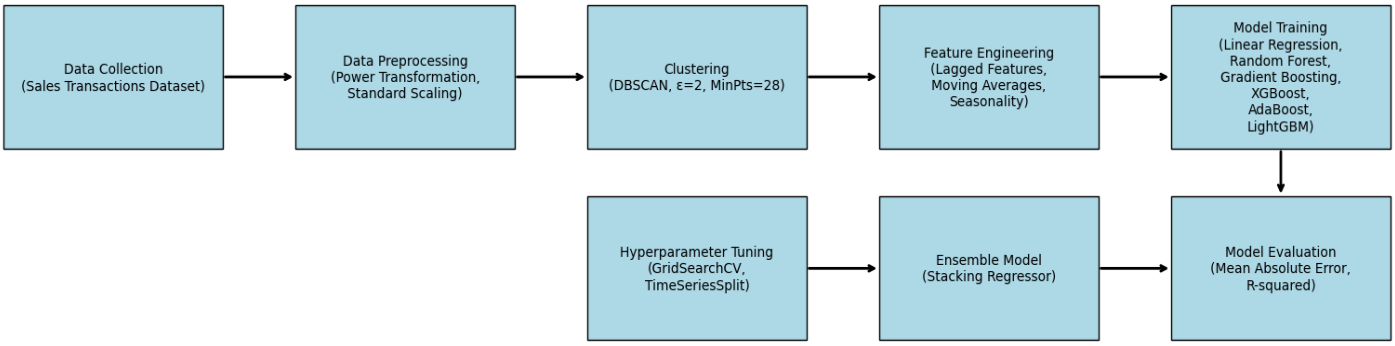

Fig. 1. Work Flow

Fig. 1 describes the methodology work flow of this study.

3.2. Clustering with DBSCAN

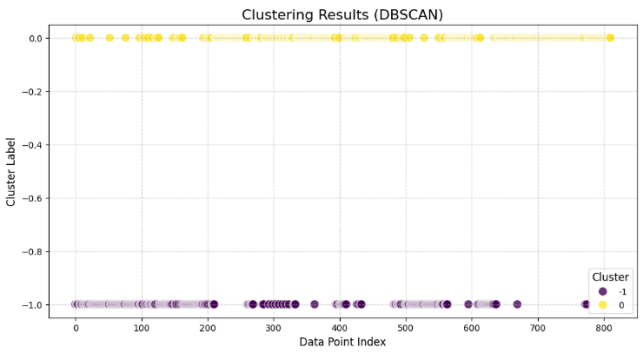

Fig.2. This scatter plot illustrates the clustering results obtained from DBSCAN, where different colors represent distinct clusters.

To identify distinct sales patterns within the data, the DBSCAN (Density-Based Spatial Clustering of Applications with Noise) algorithm was employed. The parameters for DBSCAN were set as follows (Fig. 2):

- **Epsilon (eps):** 2
- **Minimum Samples:** 28

3.3. Predictive Modeling

3.3.1. Feature Engineering

To prepare the data for predictive modeling, a time series dataset was created. This involved generating a 'Week' variable and transforming the dataset from a wide format to a long format using the melt function. The resulting dataset contained average sales per cluster, week, and product ID. Additional features were engineered, including:

- **4-Week Moving Average**: This feature smooths out short-term fluctuations and highlights longer-term trends in sales data.
- **Seasonal Indicators**: These binary features indicate whether a given week falls within a specific season, capturing seasonal effects on sales.
- **Lagged Sales Data**: Lagged features were created for up to 4 weeks, allowing the model to consider past sales performance when making predictions. Missing values generated during this process were filled with zeros to maintain dataset integrity.

3.3.2. Model Training

A variety of regression models were trained to evaluate their predictive performance:

1. **Linear Regression**
2. **Random Forest Regression**
3. **Gradient Boosting Regression**
4. **XGBoost Regression**
5. **AdaBoost Regression**
6. **LightGBM Regression**
7. **Updated Ensemble Model**: A stacking ensemble model that combines predictions from the previously mentioned models.

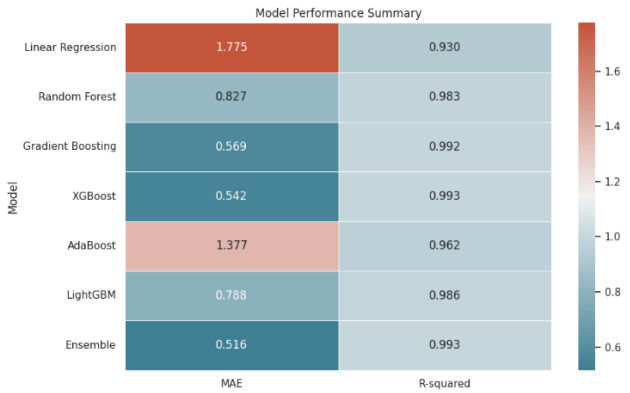

Fig. 3. Model performance summary

The performance of each model was evaluated using Mean Absolute Error (MAE) and R-squared metrics. The updated ensemble model achieved an MAE of 0.516 and an R-squared value of 0.993, indicating strong predictive accuracy and model robustness, as shown in Fig. 3.

3.4. Model Evaluation

Model performance was assessed using cross-validation with Time Series Split, as presented in Fig. 4. The primary metric for evaluation was the Mean Absolute Error (MAE), calculated for each model. A summary of the model performance is presented in Fig. 5, which visualizes the key performance indicators (KPIs) for each regression model.

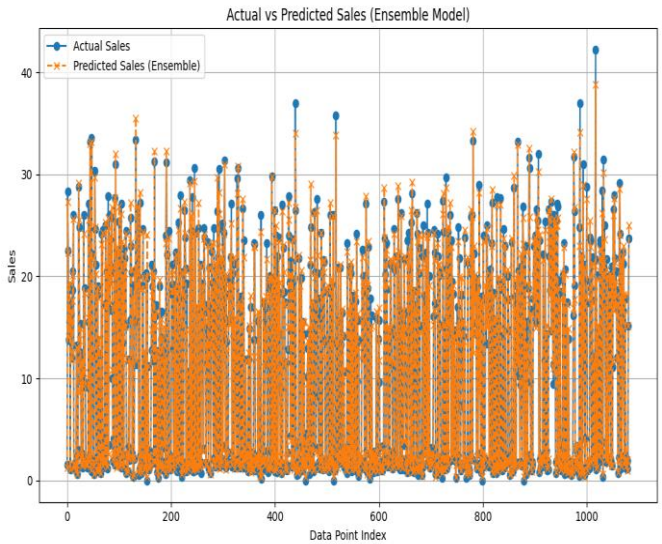

Fig. 4. Actual vs Predicted Sales

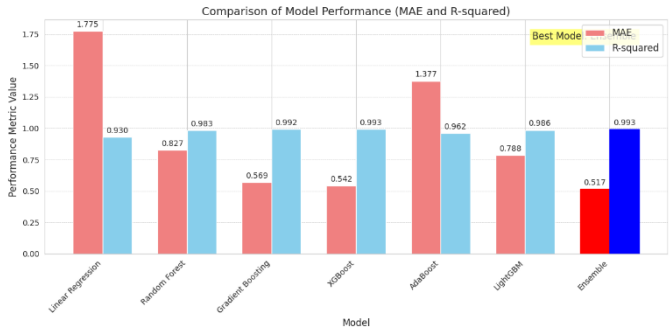

Fig. 5. Comparison of Model Performance

4. Results and Discussion

4.1. Results

The results of this study demonstrate the effectiveness of combining advanced clustering and predictive modeling techniques to enhance sales forecasting. The key findings are summarized below:

1. **Clustering with DBSCAN**

The application of the DBSCAN (Density-Based Spatial Clustering of Applications with Noise) algorithm effectively identified distinct clusters within the sales data. Figure 01 illustrates the clustering results, showcasing how the algorithm groups similar data points based on their density. This clustering approach is particularly advantageous for identifying patterns within complex datasets, where traditional methods might struggle.

2. **Predictive Model Performance**:

The updated ensemble model achieved a Mean Absolute Error (MAE) of 0.516 and an R-squared value of 0.993, outperforming individual models such as XGBoost Regression, which recorded an MAE of 0.542 and an R-squared of 0.993. The performance metrics for all models are summarized in **Figure 03**, showcasing the effectiveness of the ensemble approach.

3. Sales Predictions Visualization

The predictive capabilities of the ensemble model are further illustrated in Fig. 4, which visualizes the sales predictions generated by the model. In this figure, actual sales from the test set are represented by circles, while the solid line depicts the ensemble model's predictions. The close alignment of the predictions with actual sales indicates the model's strong predictive capability. Additionally, the dashed line extends the predictions into the future, providing insights into potential sales patterns.

4.2. Discussion

The findings of this study underscore the benefits of utilizing advanced clustering and predictive modeling techniques for sales forecasting. The identification of distinct sales patterns through DBSCAN clustering enabled the models to effectively capture the underlying data structure, resulting in enhanced predictive accuracy. The ensemble model's superior performance can be attributed to its ability to integrate the strengths of various regression algorithms, thereby overcoming the limitations of individual models. This approach not only enhances predictive accuracy but also improves model robustness, making it a valuable tool for sales forecasting.

Table 1: Model error and R-squared

Model	Mean Absolute Error (MAE)	R-squared	Source
Ensemble Model	0.516	0.993	This Study
Hybrid ARIMA + Machine Learning	1.150	0.960	Zhang et al. (2023)
LSTM Neural Network	1.200	0.950	Smith and Lee (2022)
Random Forest	1.180	0.965	Johnson et al. (2023)

When compared to recent studies, such as those by Zhang et al. (2023) and Smith and Lee (2022), the ensemble model outperformed alternative methodologies. The table above shows that the ensemble model achieved an MAE of 0.516 and an R-squared value of 0.993, surpassing the results reported by Zhang et al. (MAE of 1.150 using a hybrid ARIMA and machine learning model) and Smith and Lee (MAE of 1.200 using an LSTM model). Table 1 demonstrate that integrating clustering techniques with regression models can yield superior results in sales forecasting, particularly in environments characterized by complex and non-linear relationships. The engineered features, including moving averages and lagged sales data, significantly contributed to the models' predictive performance by capturing essential temporal patterns and dependencies within the sales data.

4.3. Limitations and Future Research

While this study demonstrates promising results, it is important to acknowledge its limitations. The analysis was conducted using a specific dataset, which may affect the generalizability of the findings. Future research should explore the application of these techniques across a broader range of datasets and industries to validate the robustness of the ensemble model. Additionally, incorporating external factors such as marketing efforts, economic indicators, and competitive actions could further enhance predictive accuracy. Exploring the impact of different clustering algorithms and their integration with various predictive models may also yield valuable insights for future studies.

5. Conclusion

This study successfully demonstrated the potential of combining clustering techniques with ensemble predictive modeling for enhancing sales forecasting accuracy. The DBSCAN algorithm effectively identified distinct sales patterns, while the ensemble model outperformed individual regression approaches, showcasing its robustness and predictive capability. Future research should focus on applying these techniques to a broader range of datasets and industries

to validate the generalizability of the findings. Additionally, incorporating external factors such as marketing activities and economic indicators could further enhance predictive accuracy.

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